Mount Kenya University

UNIVERSITY EXAMINATION 2024/2025

SCHOOL OF PURE AND APPLIED SCIENCES
DEPARTMENT OF NATURAL SCIENCES

BSCAS/BSS/BEDA/BEDS/BSNE
SCHOOL BASED/DIBEL

UNIT CODE: BMA1102 UNIT TITLE: CALCULUS I

DATE: TUE 10TH DEC, 2024 7.00AM MAIN EXAM TIME: 2 HOURS

ANSWER QUESTION ONE IN SECTION A AND ANY OTHER TWO QUESTIONS FROM SECTION B

SECTION A
QUESTION ONE (30 MARKS)

a) Evaluate the following

i) $\lim_{x \to -3} \frac{x^2 + x - 6}{x^2 - 9}$ (3 Marks)

ii) $\lim_{x \to 0} \frac{\sin 5x}{\sin 7x}$ (3 Marks)

b) Find the derivative of $y = \sqrt{5z - 8}$ by the chain rule. (4 Marks)

c) A manufacturer needs to make a cylindrical that will hold 1.5 litres of liquid. Determine the dimensions of the can that will minimize the amount of material used in its construction. (4 Marks)

d) Apply the definition of the derivative to find $f'x$ if
$y = sin^{-1} x$ (4 Marks)

e) Find the derivatives of the following function

i) $y = x^3 \, sin\, 5x$ (4 Marks)

ii) $y = \ln\left(\frac{3x^2 - 2}{3x^2 - 4}\right)$. (4 Marks)

f) Given that $x^3 + y^3 + 3xy^2 = 8$, determine value of $\frac{dy}{dx}$ at (3,2). (4 Marks)

QUESTION TWO

a) Find $\frac{dy}{dx}$ if

i) $y = \sin^2 3x \cos^4 5x$. (5 Marks)

ii) $y = (3x - 2x^2)^3$. (5 Marks)

b) Air is being pumped into a spherical balloon at a rate of 4.5 cubic feet per minute. Find the rate of change of the radius when the radius is 2 feet. (5 Marks)

c) Show that $\frac{d}{dx}(sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}} |x| > 1$. (5 Marks)

QUESTION THREE

a) If $x^2 + y^2 - 2x + 2y = 23$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at (-2,3). (5 Marks)

b) If $y = cos\, 2t$ and $x = sin\, t$, find $\frac{d^2y}{dx^2}$. (5 Marks)

c) Find the equation of tangent and normal to the curve $x^2 + y^2 + 3xy - 11 = 0$ at (1,2). (5 Marks)

d) A hyperbola is described by the equation $xy = 4$. Determine its radius of Curvature at (2,2). (5 Marks)

QUESTION FOUR

a) Find the derivative of:

i) $y = tanh^{-1}\left(\frac{3x}{4}\right)$ (5 Marks)

ii) $y = (1 - x^2)\, sin^{-1} x$ (5 Marks)

iii) $y = x^4 e^{3x}\, tan\, x$ (5 Marks)

b) A Farmer requires to fence a rectangular field. He has 500 feet of fencing wire, and a building is on one side of the field hence wont need fencing. Determine the dimensions of the field that will enclose the largest area. (5 Marks)

QUESTION FIVE

a) Find $\frac{dy}{dx}$ if $y^2 + y^3 = 4x^3$. (5 Marks)

b) The cost (in dollars) of producing x units of a given commodity is $c(x) = 5000 + 10x + 0.05x^2$.

i) Find the average rate of change of c with respect to $x$when the production level is changed from x =100 to x =105. (3 Marks)

ii) Find the instantaneous rate of change of c with respect to x (Marginal Cost) when x =100. (4 Marks)

c) A store has been selling 200DVD burners a week at Ksh.350 each. A market survey indicates that for each Ksh.10 reduction in price offered to buyers, the number of units sold will increase by 20 per week. Determine,

i) Demand function. (3 Marks)

ii) Revenue function. (2 Marks)

iii) How large a price reduction the store should offer to maximize revenue. (3 Marks)